

Lemma:

Suppose L and K are compactly generated Elementary groups and K acts on a countable set with compact / open point stabilisers (no need to be a faithful action). Let $U \in \mathcal{U}(L)$, then we have that

$$\delta + \alpha \leq \mathcal{G}(L S_0(K, X)) \leq \mathcal{G}(L) + \mathcal{G}(K)$$

where $\mathcal{G}(L) = \delta + 2$, $\mathcal{G}(K) = \alpha + 2$.

Proof:

→ Upper bound follows from $\{1\} \rightarrow \bigoplus_x (L, U) \rightarrow L S_0(K, X) \rightarrow K \rightarrow \{1\}$
and $\mathcal{G}(L) = \mathcal{G}(\bigoplus_x (L, U))$

→ Case K compact follows from the same argument as with the other lemma.

→ For the case K non-compact, we will make an induction on δ and α , one after the other

→ Base case: $\delta = 0$ follows from the fact

$$\alpha \leq \mathcal{G}(K) \leq \mathcal{G}(L S_0(K, X))$$

→ both non-Trivial

→ Suppose that for all Q compactly generated Elementary group with $\mathcal{G}(Q) < \delta$ and for all E Elementary group with $\mathcal{G}(Q) = \delta + 2$ and $\mathcal{G}(E) = \beta + 2$

compactly generated

↳ acting on X countable

we have, for $U \in \mathcal{U}(Q)$

$$\delta + \beta < \mathcal{G}(Q S_0(E, X))$$

→ Let L be s.t. $\mathcal{G}(L) = \delta + 2$ and K Elementary and compactly generated, $\mathcal{G}(K) = \alpha + 2$

→ Because K is non-compact, let $Z \subset X$ be an infinite orbit

→ Let $U \in \mathcal{U}(L)$ and define

$$N := \langle\langle U \rangle\rangle_L, \quad H := N \dot{\cup}_0 (K, Z), \quad J := \bigoplus_Z (N, U)$$

observe that H is closed in $L \dot{\cup}_0 (K, X)$, then:

$$\mathcal{G}(H) \leq \mathcal{G}(L \dot{\cup}_0 (K, X))$$

also, because $N \dot{\cup}_0 L$ then $\text{Res}(L) \leq N$ and, because L is compactly generated

$$\delta + 1 = \mathcal{G}(\text{Res}(L)) \leq \mathcal{G}(N) \leq \delta + 2 = \mathcal{G}(L)$$

Let $(Q_i)_{i \in \mathbb{N}}$ be an increasing family of compactly generated subgroups with $\bigcup_i Q_i = N$. Then $\sup_{i \in \mathbb{N}} \mathcal{G}(Q_i) \leq \delta + 1$. → we also want $U \leq Q_i \forall i$

define $H_i := Q_i \dot{\cup}_0 (K, Z)$ and

By the induction hypothesis we have $(\mathcal{G}(Q_i) = \beta_i + 2 \leq \delta + 1 \leq \mathcal{G}(N))$
 $\beta_i + \alpha \leq \mathcal{G}(H_i)$ → implies $\beta_i < \delta$

Then

$$\begin{aligned} \sup \beta_i + \alpha &\leq \mathcal{G}(H) \\ \delta + \alpha &\leq \mathcal{G}(H) \leq \mathcal{G}(L \dot{\cup}_0 (K, X)) \end{aligned}$$

as desired. ■